

# Player positions and the 4×4 Q matrix

We examine twelve representative soccer states. For each, the **left panel** shows the physical player configuration and every action either player actually takes in equilibrium, drawn as probability arrows (thick = likely, thin = unlikely, no arrow at all = zero probability). The **right panel** shows the complete 4×4 stage-game Q matrix — the payoff table  $Q(s, a_0, a_1)$  at that one state, not the value function  $V(s)$  — redrawn as a node-and-arrow graph. Always the full grid, never reduced. The goal is not simply to show *where* mixing happens on the board, but **what the players are actually doing when it does**.

A player's **support** is the set of actions it assigns positive probability in an equilibrium strategy — support (2,2) means the carrier mixes over 2 actions and the defender mixes over 2 actions.

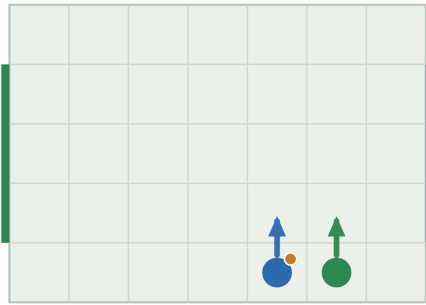
## Canonical support shapes

| CARRIER SUPPORT    | DEFENDER SUPPORT | SHAPE | STATES |
|--------------------|------------------|-------|--------|
| 2 actions          | 2 actions        | (2,2) | 68     |
| 3 actions          | 3 actions        | (3,3) | 4      |
| 2 actions          | 1 action         | (2,1) | 14     |
| 3 actions          | 2 actions        | (3,2) | 8      |
| total mixed states |                  |       | 94     |

The defender's support is never larger than the carrier's, in any of the 94 mixed states on the canonical board.

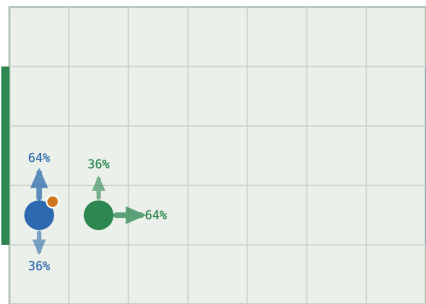
**The bigger finding underneath those four rows:** 94 mixed states are not 94 different phenomena — they are four repeated shapes, recurring all over the board. Case 7 shows one of those repeats directly (Case 2's exact shape, elsewhere on the board, mirrored). This page shows one example of each of the four shapes, plus every distinct mechanism studied in the project that can force a mix: a stochastic move order, this project's own tackle rule, a dense reward with zero transition noise, and movement slip on a goal shape that is otherwise always pure.

### A pure state -- one safe cell, nothing to guess



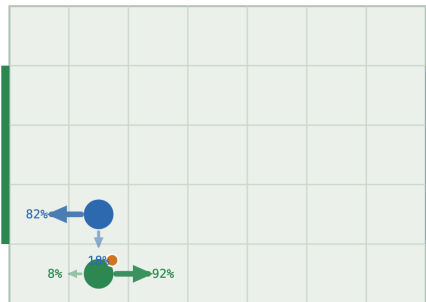
pure equilibrium ·  $V = +0.234$

### The primary example: the typical mix



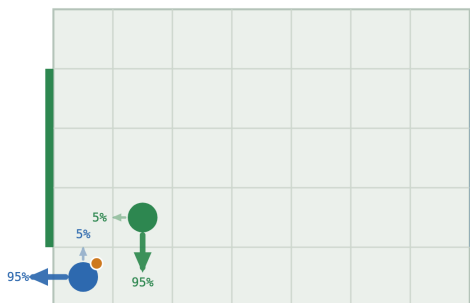
mixed equilibrium ·  $V = +0.094$

### A clean L/R indifference example



mixed equilibrium ·  $V = -0.206$

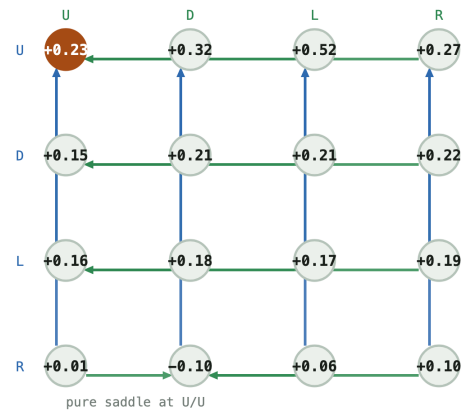
### The corner duel -- (0,0) and (1,1)



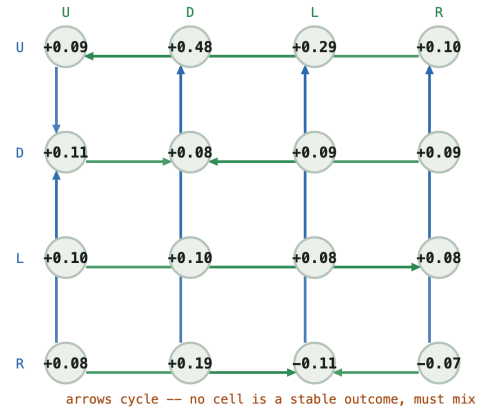
mixed equilibrium ·  $V = +0.078$

### A genuine 3-action mix, not a 2-cycle

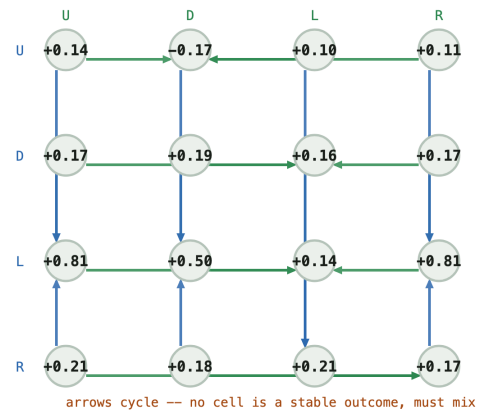
### Q matrix -- (4, 0, 5, 0, 0)



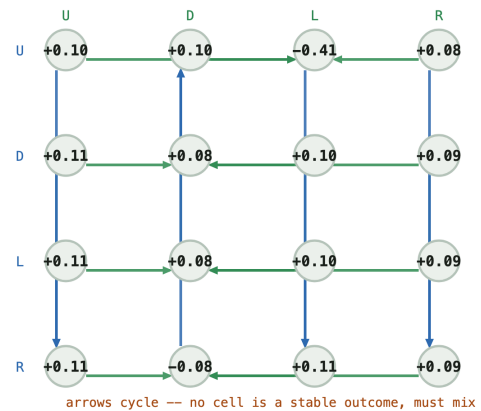
### Q matrix -- (0, 1, 1, 1, 0)



### Q matrix -- (1, 1, 1, 0, 1)

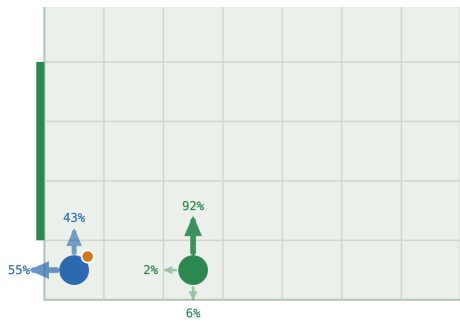


### Q matrix -- (0, 0, 1, 1, 0)



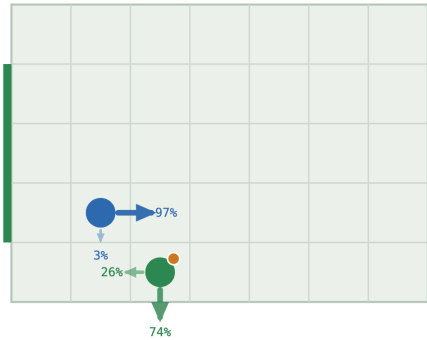
### Q matrix -- (0, 0, 2, 0, 0)





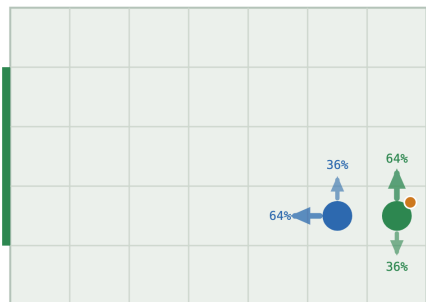
mixed equilibrium ·  $V = +0.085$

**A near-pure hedge -- where rounding would lie**



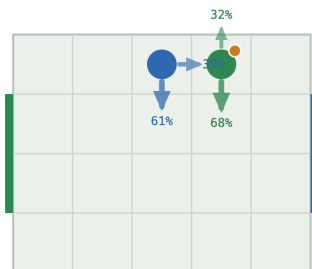
mixed equilibrium ·  $V = -0.168$

**The typical mix, mirrored -- same shape, value negated**



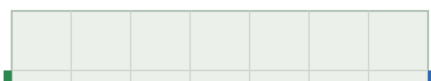
mixed equilibrium ·  $V = -0.094$

**This project's own tackle rule -- a different mechanism, same duel**



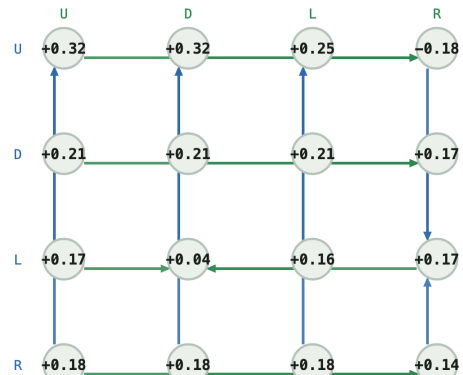
mixed equilibrium ·  $V = +0.013$

**The asymmetric mix -- a tie, not a forced mix, dressed the same**



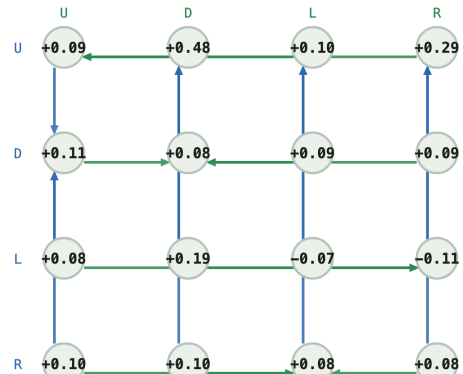
arrows cycle -- no cell is a stable outcome, must mix

**Q matrix -- (1, 1, 2, 0, 1)**



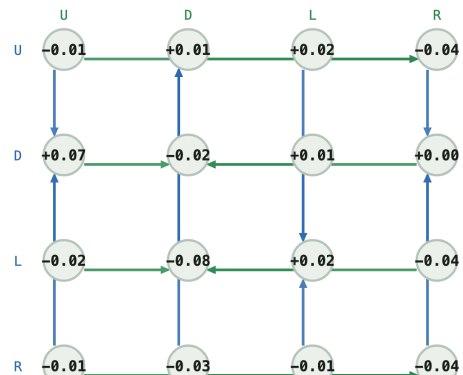
arrows cycle -- no cell is a stable outcome, must mix

**Q matrix -- (5, 1, 6, 1, 1)**



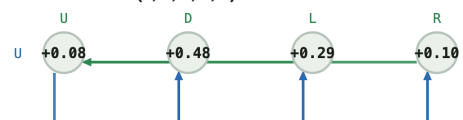
arrows cycle -- no cell is a stable outcome, must mix

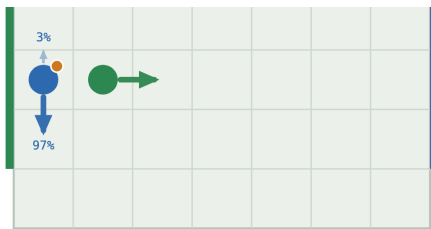
**Q matrix -- (2, 3, 3, 3, 1)**



arrows cycle -- no cell is a stable outcome, must mix

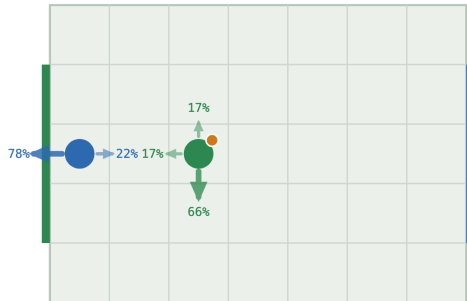
**Q matrix -- (0, 2, 1, 2, 0)**





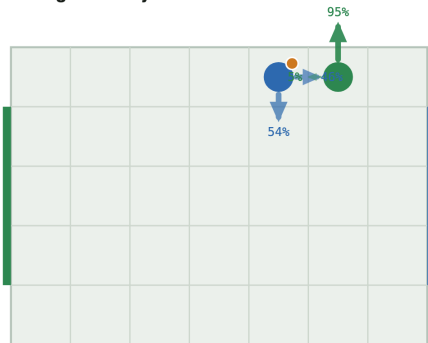
mixed equilibrium ·  $V = +0.095$

**The three-lane mix -- support (3,2), the deepest gap on this page**



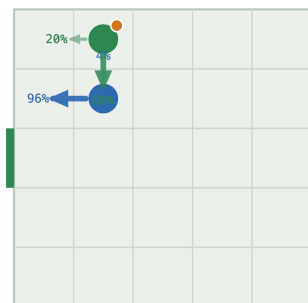
mixed equilibrium ·  $V = -0.282$

**Mixing forced by reward alone -- zero transition randomness**

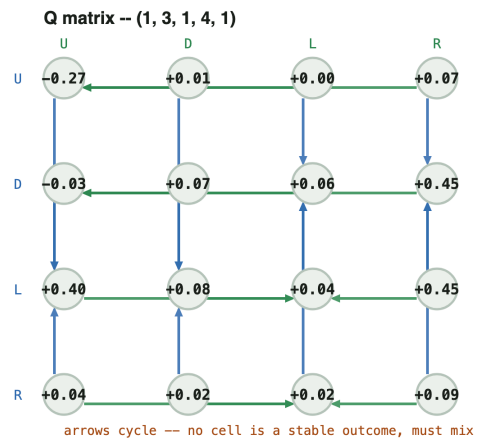
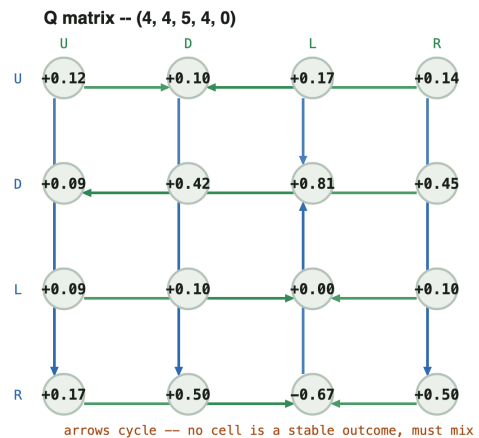
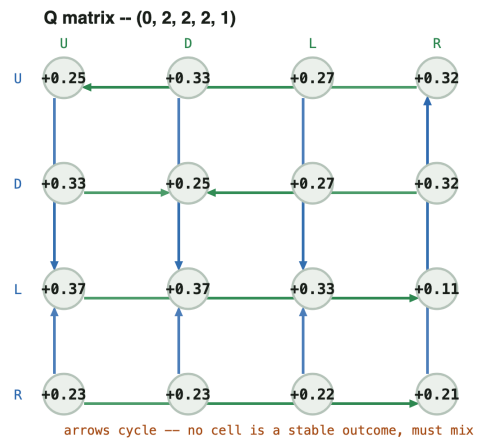
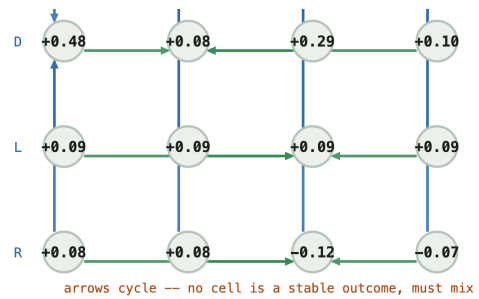


mixed equilibrium ·  $V = +0.129$

**Movement slip -- a single-cell goal, pure until now**



mixed equilibrium ·  $V = -0.053$



All twelve cases: board position with probability arrows (left) and the full 4×4 Q matrix redrawn as a node-and-arrow graph (right). A lit node with no outgoing arrow is a pure saddle; a closed best-response cycle means no pure action pair is stable.

## Reading the board and the Q matrix

**The board (left).** Each player is a filled circle; the ball sits on whoever carries it. An arrow points from a player toward a cell it might move to, and its **thickness and opacity are the equilibrium probability** of that action — a bold arrow is very likely, a thin one is unlikely, and an action with (numerically) zero probability draws no arrow at all. A pure policy is one bold arrow and nothing else; a mixed policy fans out into two or three arrows of different weights.

**The Q matrix (right).** Each node is one cell of the complete  $4 \times 4$  stage matrix (row = the carrier's action, col = the defender's action, always reoriented so the printed payoff is the carrier's, regardless of which player id actually carries the ball). A blue arrow points from a cell to the cell in the same column with the higher payoff — the carrier's own reason to switch rows. A green arrow points from a cell to the cell in the same row with the *lower* payoff — the defender's reason to switch columns, since the defender minimises. Every case shows all 16 cells, so the shapes are directly comparable. This is the mathematical justification *for* what the board already shows: it is why the players' support is exactly what it is.

A pure state has exactly one node with no outgoing arrow — that cell is the saddle, everything else eventually points to it. A mixed state has arrows everywhere: every node points somewhere, so they chase each other around a closed best-response cycle — no pure action pair is stable, which is exactly why the board shows more than one arrow.

## Case 1 (4, 0, 5, 0, 0)

A pure state, for contrast. Saddle at U/U. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

**Support:** carrier {U} (100%) · defender {U} (100%)

|   | U     | D      | L     | R     |
|---|-------|--------|-------|-------|
| U | 0.234 | 0.317  | 0.523 | 0.272 |
| D | 0.151 | 0.211  | 0.211 | 0.222 |
| L | 0.161 | 0.182  | 0.172 | 0.190 |
| R | 0.012 | -0.095 | 0.058 | 0.098 |

## Case 2 (0, 1, 1, 1, 0)

The primary example — the typical mix. Gap 0.0136. The carrier mixes U 63.5% / D 36.5%; the defender answers U 36.5% / R 63.5%. This is not one strange state among many — it is **the dominant pattern in the entire mixed-state set: 90 of the 94 mixed states on the canonical board cross a vertical pair exactly like this one**, the carrier deciding which goal row to attack and the defender guessing which one. Everything that follows is either this same shape recurring (Case 7), or a genuinely different shape worth contrasting against it (Cases 3, 4, 5, 9, 10).

**Support:** carrier {U, D} (63.5% / 36.5%) · defender {U, R} (36.5% / 63.5%)

|   | U     | D     | L      | R      |
|---|-------|-------|--------|--------|
| U | 0.086 | 0.478 | 0.286  | 0.099  |
| D | 0.109 | 0.076 | 0.085  | 0.086  |
| L | 0.103 | 0.103 | 0.085  | 0.084  |
| R | 0.082 | 0.187 | -0.108 | -0.071 |

### Case 3 (1, 1, 1, 0, 1)

A clean L/R indifference example. Placed directly after the primary example on purpose: this is the cleanest demonstration of the intuition Case 2 only implies —

*"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."*

Gap 0.0445. The carrier's entire live option set is exactly  $\{L, R\}$  — no vertical option survives at all, which is what makes this state (rather than Case 4, which crosses U/L) the cleanest match to that description: two actions genuinely equally good against the defender's own mix. **This is the rarer shape overall: only 4 of 94 mixed states cross a purely horizontal pair instead of a vertical one.**

**Support:** carrier  $\{L, R\}$  (7.7% / 92.3%) · defender  $\{D, L\}$  (18% / 82%)

|   | U     | D      | L     | R     |
|---|-------|--------|-------|-------|
| U | 0.139 | -0.173 | 0.099 | 0.110 |
| D | 0.171 | 0.185  | 0.157 | 0.171 |
| L | 0.810 | 0.500  | 0.141 | 0.810 |
| R | 0.211 | 0.181  | 0.211 | 0.167 |



## Case 4 (0, 0, 1, 1, 0)

**The corner duel.** The board position sketched at the meeting: the carrier at (0, 0) — the back corner, right against its own goal — and the defender diagonally adjacent at (1, 1). This is the closest match, coordinate for coordinate, to that sketch. Distinct from both cases before it — Case 2 crosses U/D, Case 3 a clean L/R — this one crosses U/L for the carrier and D/L for the defender: pinned in the corner, the carrier's two live escapes are straight up the sideline (U) or across along the back line (L), and the defender is guessing which. **Gap 0.0121; the mix is lopsided but genuine** — the corner leaves little room, but not zero.

**Support:** carrier {U, L} (4.6% / 95.4%) · defender {D, L} (94.9% / 5.1%)

|   | U     | D      | L      | R     |
|---|-------|--------|--------|-------|
| U | 0.103 | 0.103  | -0.408 | 0.084 |
| D | 0.109 | 0.076  | 0.101  | 0.086 |
| L | 0.109 | 0.076  | 0.101  | 0.086 |
| R | 0.112 | -0.075 | 0.112  | 0.088 |

## Case 5

(0, 0, 2, 0, 0)

A genuine 3-action mix. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two, so three actions are simultaneously undominated on both sides. **This is one of only 4 states on the whole canonical board with support shape (3,3)** — mixing is not always a 50/50 split between two actions. (Entropy 1.109 bits, the richest mix on this page, if a single summary number is wanted — but the three-way split itself is the point.)

**Support:** carrier {U, L, R} (43.3% / 54.7% / 1.9%) · defender {U, D, L} (91.9% / 5.9% / 2.3%)

|   | U     | D     | L      | R     |
|---|-------|-------|--------|-------|
| U | 0.084 | 0.095 | 0.103  | 0.110 |
| D | 0.086 | 0.076 | 0.076  | 0.086 |
| L | 0.086 | 0.076 | 0.076  | 0.086 |
| R | 0.088 | 0.095 | -0.088 | 0.110 |

## Case 6

(1, 1, 2, 0, 1)

A near-pure hedge, where rounding would lie. Gap 0.0043. The defender plays R 97.5% / D 2.5% — entropy only 0.169 bits, a hedge so lopsided that reading 0.975 as 1.0 would be a real rounding mistake: the gap is certified and real, twelve times smaller than a 0.1 rounding grid would resolve. The carrier's own mix is more balanced (D 73.9% / L 26.1%) — **it is the defender's near-pure hedge that makes this case worth including, not the carrier's.**

**Support:** carrier {D, L} (73.9% / 26.1%) · defender {D, R} (2.5% / 97.5%)

|   | U     | D     | L     | R      |
|---|-------|-------|-------|--------|
| U | 0.317 | 0.317 | 0.247 | -0.178 |
| D | 0.211 | 0.211 | 0.211 | 0.167  |
| L | 0.171 | 0.045 | 0.157 | 0.171  |
| R | 0.182 | 0.182 | 0.182 | 0.135  |

## Case 7 (5, 1, 6, 1, 1)

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

```
V(case 2) = +0.094235
V(mirror) = -0.094235
sum       = -1.39e-17 (zero, to floating-point noise)
```

**Direct evidence for the repetition claim above:** this is not a new shape, it is Case 2's identical mix reappearing, exactly mirrored, wherever the same relative configuration recurs.

**Support:** carrier {U, D} (63.5% / 36.5%) · defender {U, L} (36.5% / 63.5%)

|   | U     | D     | L      | R      |
|---|-------|-------|--------|--------|
| U | 0.086 | 0.478 | 0.099  | 0.286  |
| D | 0.109 | 0.076 | 0.086  | 0.085  |
| L | 0.082 | 0.187 | -0.071 | -0.108 |
| R | 0.103 | 0.103 | 0.084  | 0.085  |

## Case 8

(2, 3, 3, 3, 1), 5×4 board

This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. **It produces the same kind of duel** (gap 0.0223): the carrier mixes U 32.1% / D 67.9%, the defender mixes D 60.7% / R 39.3%. Different cause, same structure.

**Support:** carrier {U, D} (32.1% / 67.9%) · defender {D, R} (60.7% / 39.3%)

|   | U      | D      | L      | R      |
|---|--------|--------|--------|--------|
| U | -0.012 | 0.005  | 0.016  | -0.042 |
| D | 0.072  | -0.022 | 0.009  | 0.000  |
| L | -0.019 | -0.080 | 0.019  | -0.042 |
| R | -0.007 | -0.026 | -0.006 | -0.039 |

## Case 9

(0, 2, 1, 2, 0)

The asymmetric mix — the case worth discussing most. Support (2, 1) — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (U 2.6% / D 97.4%), but the **defender's** equilibrium is a single fixed move (R, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

**Support:** carrier {U, D} (2.6% / 97.4%) · defender {R} (100%)

|   | U     | D     | L      | R      |
|---|-------|-------|--------|--------|
| U | 0.085 | 0.478 | 0.291  | 0.095  |
| D | 0.478 | 0.085 | 0.291  | 0.095  |
| L | 0.093 | 0.093 | 0.086  | 0.093  |
| R | 0.082 | 0.082 | -0.122 | -0.072 |

Why the carrier still needs two actions against a defender who never varies: against the defender's pure R, column R reads U 0.095, D 0.095, L 0.093, R -0.072 — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 3's 7.7% / 92.3% is. This is the sharpest example on this page of a general fact worth stating plainly: **a fractional LP output is not automatically a strategically required mixed strategy**. A tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

## Case 10

(0, 2, 2, 2, 1)

The three-lane mix. Support (3, 2) — gap 0.0699, **the deepest gap of any state on this page** (fifteen times deeper than the shallowest hedge in Case 6). This is the fourth and last canonical support shape: the carrier genuinely needs three live actions, while the defender only ever needs two. Combined with Case 9, this settles a structural question the certificates make checkable rather than assumed: **the defender never needs strictly more actions than the carrier**, across all 94 mixed states on the canonical board.

**Support:** carrier {U, D, L} (16.9% / 66% / 17.1%) · defender {L, R} (77.9% / 22.1%)

|   | U     | D     | L     | R     |
|---|-------|-------|-------|-------|
| U | 0.247 | 0.330 | 0.272 | 0.317 |
| D | 0.330 | 0.247 | 0.272 | 0.317 |
| L | 0.366 | 0.366 | 0.330 | 0.110 |
| R | 0.228 | 0.228 | 0.220 | 0.205 |

## Case 11 (4, 4, 5, 4, 0), deterministic

**Mixing forced by reward alone.** An advanced case, deliberately placed after the ten core examples above. Every case so far gets its coin flip from a stochastic transition — the random move order, or the tackle rule's own coin. This one has **no transition stochasticity at all**: the mix comes entirely from `scoring="territory"`, a dense per-step reward for the ball in the opponent's final third, layered on top of the ordinary win/loss score. Gap 0.0653 — close to a fair coin (entropy 0.996 bits), forced purely by coupling the reward to both players' actions. **Mixing does not require stochastic transitions; it can come from the payoff structure alone.**

**Support:** carrier {D, R} (53.8% / 46.2%) · defender {U, L} (95.1% / 4.9%)

|   | U     | D     | L      | R     |
|---|-------|-------|--------|-------|
| U | 0.116 | 0.105 | 0.170  | 0.136 |
| D | 0.094 | 0.415 | 0.815  | 0.450 |
| L | 0.094 | 0.105 | 0.000  | 0.105 |
| R | 0.170 | 0.500 | -0.671 | 0.500 |



## Case 12 (1, 3, 1, 4, 1)

**Movement slip on a single-cell goal.** The sharpest possible contrast with Case 1. A single-cell goal under deterministic move order is **pure across every configuration tested in this project** — that is the whole goal-width result; a board-size-free proof remains open. Add `slip=0.15` (each player independently takes a uniform-random move instead of its chosen one, 15% of the time, regardless of position) to that exact board, and **52 mixed states appear where there were 0**. Gap 0.0172 — a genuine, certified mix, not noise. The goal-width switch is a property of Littman's move-order coin specifically; a coin that lands on every square regardless of the goal forces mixing everywhere, independent of goal width.

**Support:** carrier {D, L} (80.2% / 19.8%) · defender {U, L} (3.9% / 96.1%)

|   | U      | D     | L     | R     |
|---|--------|-------|-------|-------|
| U | -0.273 | 0.006 | 0.001 | 0.069 |
| D | -0.031 | 0.069 | 0.057 | 0.450 |
| L | 0.396  | 0.083 | 0.039 | 0.450 |
| R | 0.039  | 0.020 | 0.015 | 0.092 |

## The mathematics behind all twelve

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own mix — that equality is what "indifferent" means in Case 3, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different explanations. Case 9 shows the boundary of that story: indifference alone (a tie against a fixed opponent) can produce the same *symptom* — a fractional policy — without a cycle anywhere in the matrix.

Reproduce with `python scripts/positions.py (make positions)` — writes `figures/gallery/positions.svg` and prints every matrix, policy, and support shown here. Full project: [github.com/Charith-Reddy-Pareddy/soccer-markov-nash](https://github.com/Charith-Reddy-Pareddy/soccer-markov-nash).